

Two players take turns taking counters. Unless a problem says something else, a player takes 1 or 2 counters on each turn, and whoever takes the last counter wins.

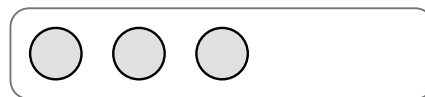
Problem 1: On your turn, circle the counters you take to be sure to win. Cross out each pile where you cannot be sure to win.



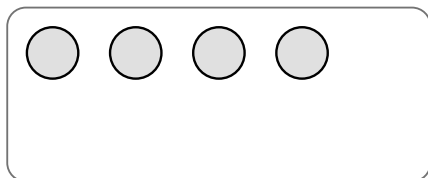
1



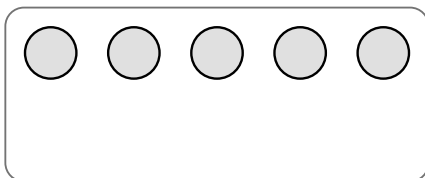
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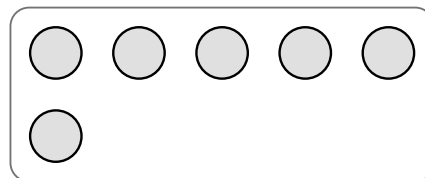
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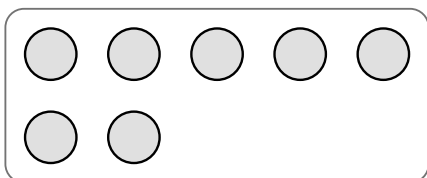


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6

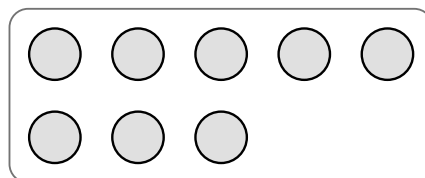
Problem 2: Would you rather go 1st or 2nd with each pile? Circle 1st or 2nd.



7

1st

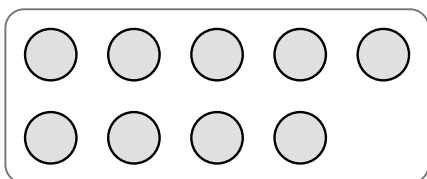
2nd



8

1st

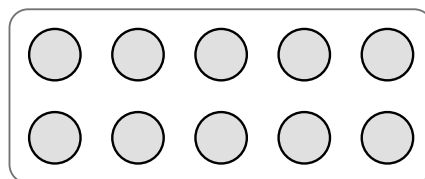
2nd



9

1st

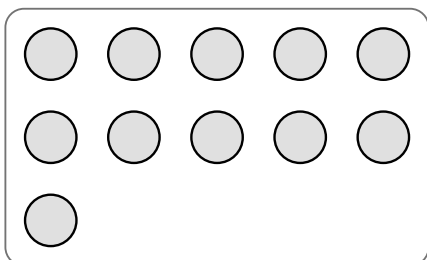
2nd



10

1st

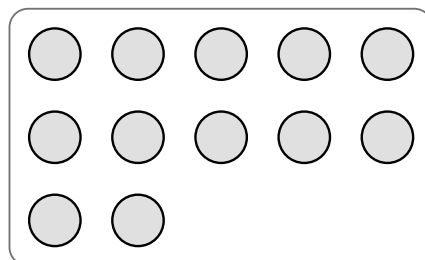
2nd



11

1st

2nd



12

1st

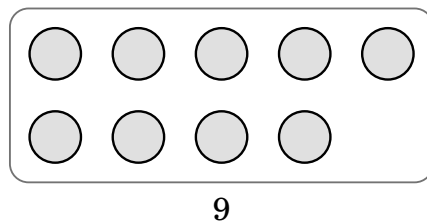
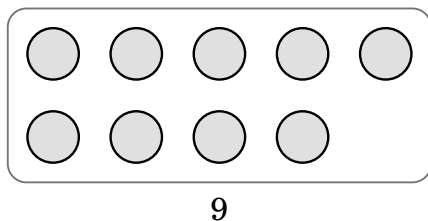
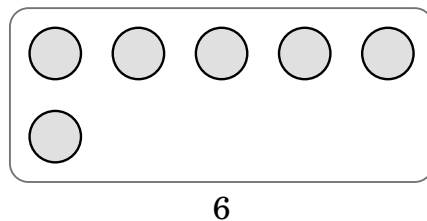
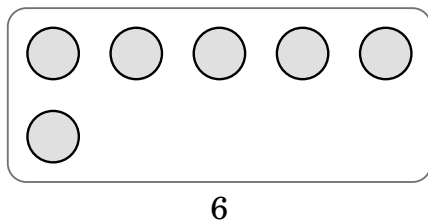
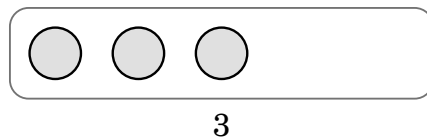
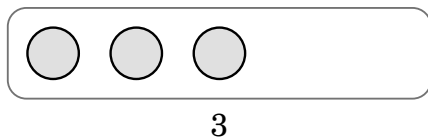
2nd

Problem 3: A trap is a pile you do not want to have on your turn. Put an X on every trap from 1 to 20.

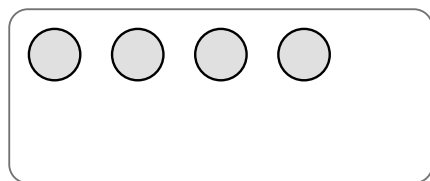


1 •	2 ••	3 •••	4 ••••	5 •••••	6 ••••• •	7 ••••• ••	8 ••••• •••	9 ••••• ••••	10 ••••• •••••
11 ••••• •	12 ••••• ••	13 ••••• •••	14 ••••• ••••	15 ••••• •••••	16 ••••• ••••• •	17 ••••• ••••• ••	18 ••••• ••••• •••	19 ••••• ••••• ••••	20 ••••• ••••• •••••

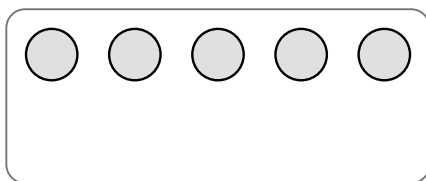
Problem 4: It is the other player's turn. For each move they can make, cross out what they take, then circle what you take to win.



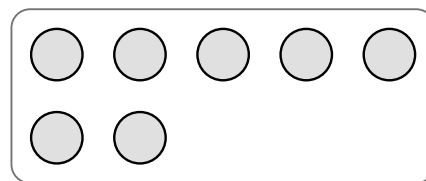
Problem 5: Ben always starts by taking 2 counters. Circle each pile where Ben can still be sure to win after that.



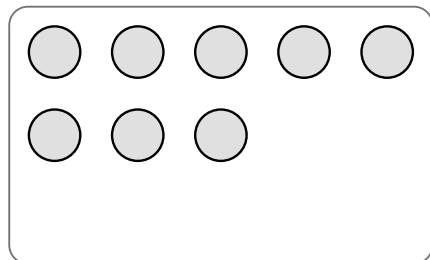
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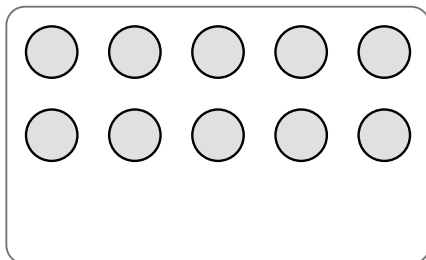
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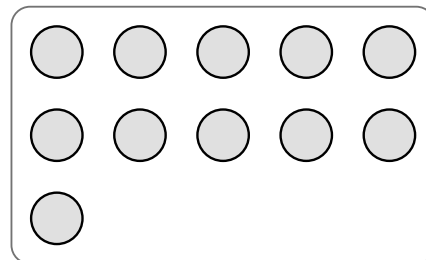
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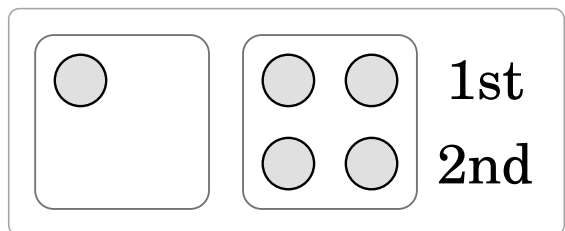
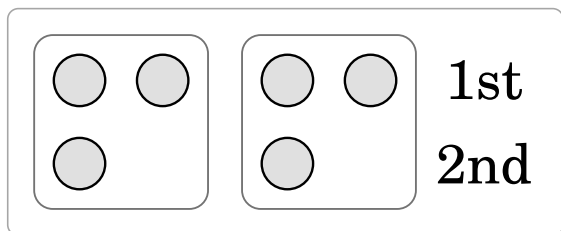
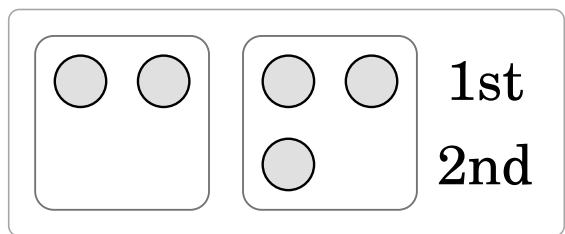
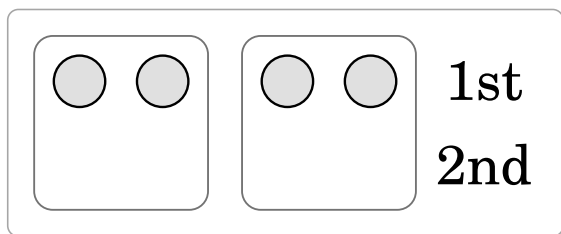
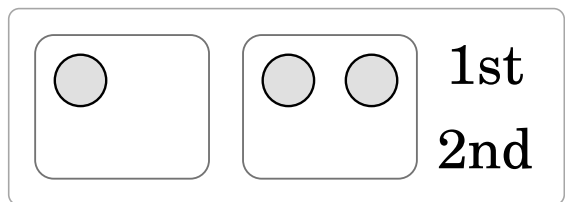
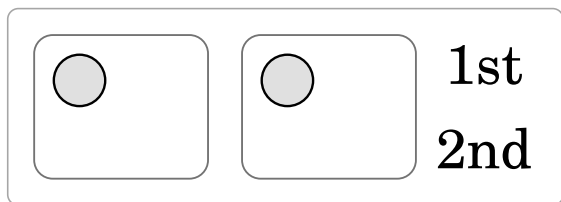
11

Problem 6: Now each player may take 1, 2, or 3 counters on a turn. Put an X on every trap from 1 to 20.

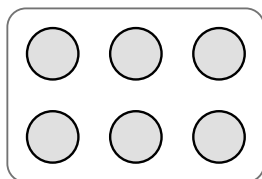
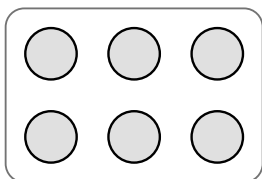


1 •	2 ••	3 •••	4 ••••	5 •••••	6 ••••••	7 •••••••	8 ••••••••	9 •••••••••	10 ••••••••••
11 ••••••••••	12 ••••••••••••	13 •••••••••••••	14 ••••••••••••••	15 •••••••••••••••	16 ••••••••••••••••	17 •••••••••••••••••	18 ••••••••••~••••••••	19 ••••••~••••••••••••	20 ••••••••••••••••••

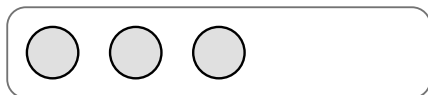
Problem 7: Now there are two piles, and on your turn you take 1 or 2 counters from one pile. Would you rather go 1st or 2nd? Circle it.



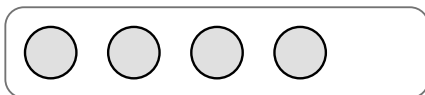
Problem 8: Play the same game with two piles of 6, and go 2nd. Find a way to win every time, and show the adult.



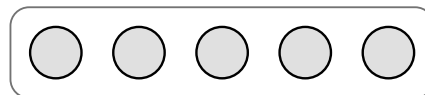
Problem 9: Now there is one pile, and each player may take 1 or 3 counters. Circle what you take to be sure to win, and cross out each trap.



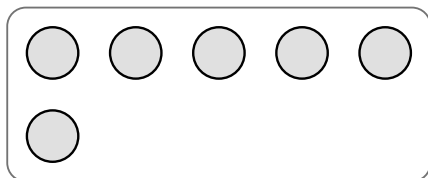
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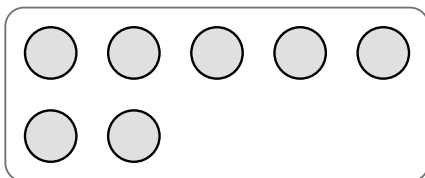
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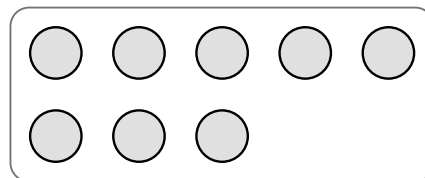
5



6



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Problem 10: Now each player takes 1 or 2 counters again, and whoever takes the last counter loses. Put an X on every trap from 1 to 20.



1 •	2 ••	3 •••	4 ••••	5 •••••	6 ••••• •	7 ••••• ••	8 ••••• •••	9 ••••• ••••	10 ••••• •••••
11 ••••• •	12 ••••• ••	13 ••••• •••	14 ••••• ••••	15 ••••• •••••	16 ••••• ••••• •	17 ••••• ••••• ••	18 ••••• ••••• •••	19 ••••• ••••• ••••	20 ••••• ••••• •••••